C1 · Component sheet

Every component with its states. Class names are in sonariq.css. Panel components use plain Qt widgets: QLabel, QPushButton, QListWidget, QTextBrowser, QLineEdit.

dark primary · light shown where it prints

STATUS PILL · 7 STATES · .SQ-PILL.S-*

Fail s-fail · Warning s-warn · Pass s-pass · Accepted with reason s-acc · Cannot verify s-unk · Dormant s-dor · Not applicable s-na

Fail · Warning · Pass · Accepted with reason · Cannot verify · Dormant · Not applicable

Fail · Pass · Cannot verify lg · inline · icon 14 px

CHIPS · DISCIPLINE · LOCATION

FS Fire & life safety · ST Structure · AR Architecture · HV HVAC · SY Systems · AC Accessibility

Deck 5 · Fr 165 · P · Deck 5 · Fr 152 · S · Deck 5 · Fr 146–150 · P · Stair 3B · Decks 4–7 · MVZ 4/5 · Deck 5 · CL

Passenger ship · > 36 passengers · ~~Class · DNV~~ profile · pack off

RULE BADGE · FITS LONG REFS · .SQ-RULE

SOLAS II-2/9.2.2.1.1

SOLAS II-2/13.3.1.2

FSS Code Ch.13 §2.1.2

ADA 2010 §404.2.3

PVAG 2013 draft · corridor clear width

DNV Pt.6 Ch.1 Sec.4 [2.3.1] · sample long reference wraps on any character

Rule engine · ✓ AI explanation · numbers checked

MEASUREMENT · UNIT TOGGLE PER VALUE

1,350 mm	default · mm
30 in · 762 mm	rule in inches (ADA) + alt
915 mm · 36 in	hover · shows alt unit
4.2 m	lengths over 1 m · m

FORMULA BLOCK · INPUTS ARE CLICKABLE

$$W = (N_1 + N_2 + 0.5\,N_3 + 0.25\,N_4) \times 10 \text{ mm}$$
$$= (\,68_{N_1\ D5} + 44_{N_2\ D6} + 0.5 \cdot 40_{N_3\ D4} + 0.25 \cdot 24_{N_4\ D7}\,) \times 10$$
$$= 138 \times 10 = \mathbf{1{,}380 \text{ mm}}$$

Input states: default · hover/selected (accent outline; spaces counted are highlighted in 3D).

REQUIRED VS MEASURED

REQUIRED	MEASURED	MARGIN
1,380 mm	1,350 mm	−30 mm

FINDING ROW · DEFAULT · HOVER · SELECTED · CHANGED

Fail F-01 Fire door class below bulkhead — Deck 5 · Fr 165 · P — SOLAS II-2/9.2.2.1.1

Warning W-01 Corridor below width guideline — Deck 5 · Fr 152 · S — PVAG 2013 draft

Cannot verify U-01 Door has no fire class — D-5-171-S add the property or confirm

Fail F-07 Stair 3B too narrow changed — Stair 3B · Decks 4–7 — FSS Code Ch.13 §2.1.2

Accepted F-08 Dead-end corridor — Deck 5 · Fr 146–150 · P — approved by Fire & life safety lead

Rows 1–5: default · hover · selected (accent outline) · changed after ping (tag + 900 ms tint) · accepted.

SUMMARY STRIP · CLICK TO FILTER

1 Fail	1 Warning	1 Cannot verify	89 Pass	4 Dormant

Second cell: active filter. After a ping, changed counts turn accent for 2 s.

EDIT PREVIEW CARD · AI DRAFTS, ENGINEER CONFIRMS

EDIT PREVIEW — OR-117

Action	Add cabin block × 4
Where	Deck 5 · aft of Fr 150
Berths	8
Change order	OR-117

Confirm · Cancel · Nothing changes until you confirm.

Confirmed 14:32 · Rev B · Cancelled: card collapses to one line, model untouched

PING CARD · EACH ROW CLICKABLE

14 checks re-run · 3 changed — 0.4 s

Fail F-07 Stair 3B — Pass → Fail

Fail F-08 Dead-end corridor — Pass → Fail

Fail F-09 Corridor pinch — Pass → Fail

FIX CARD · VERIFIED · APPLIED · REJECTED

2 · Keep four cabins, one berth each — Pass

N_1 64 → 1,340 ≤ 1,350 mm · margin 10 mm

Apply · Preview in 3D · Re-checked

2 · Keep four cabins, one berth each — Applied

Applied 14:36 · Rev B · Undo available until the next save

▸ 1 idea rejected by the engine

Move the block 10 frames forward → still 1,380 mm

CHAT MESSAGES · USER · AI EXPLANATION · SYSTEM

YOU

Why does 3B fail, and what can I do?

✓ AI explanation · numbers checked

Stair 3B needs 1,380 mm and has 1,350 mm. The four new cabins raised N_1 on Deck 5 from 60 to 68.

Rev B · 14:32 · OR-117 confirmed

SIGNED HISTORY ROW

Rule engine found F-07 · 1,380 mm required, 1,350 mm measured 14:32 Rev B

Aino Laine · Outfitting designer applied fix 2 14:36 Rev B

Mikko Saari · Fire & life safety lead approved 15:10 Rev B

Hollow dot: system event. Filled dot: human, signed. Names are sample data.

BUTTONS

Analyse · Send to review · Zoom to · Apply · Disabled · Focus · ⚙

Analyse · 96 checks

INPUTS

Ask a question or describe a change

Add four cabins aft of frame 150

Reason required

Write a reason before sending for approval.

Read-only for reviewers

TABS · TOGGLES · CHECKBOXES

Findings 3 · Chat · History

Live · Live · Selection · Deck 5 · Whole model · mm · in

Fire & life safety · Class · DNV

3D PINS + FIRE-DIVISION LEGEND

F-01 Fail · W-01 Warning · U-01 Cannot verify · F-07 Pass · F-08 Accepted

selected · warning · cannot verify · pass · accepted

A-60 · A-0 · B-15 · B-0

EVIDENCE CARD · DARK (PANEL) AND LIGHT (PORTAL + PRINT)

Fail F-07 — Rule engine

Stair 3B is narrower than the people it serves need

Stair 3B · Decks 4–7 · Deck 5 · FS Fire & life safety

FSS Code Ch.13 §2.1.2 sample rule text

> Stairway clear width shall not be less than W, worked out from the persons served on each deck the stair connects.

REQUIRED	MEASURED	MARGIN
1,380 mm	1,350 mm	−30 mm

FORMULA · INPUTS FROM THE MODEL

$$W = (N_1 + N_2 + 0.5\,N_3 + 0.25\,N_4) \times 10 \text{ mm}$$
$$= (\,68_{N_1\ D5} + 44_{N_2\ D6} + 0.5 \cdot 40_{N_3\ D4} + 0.25 \cdot 24_{N_4\ D7}\,) \times 10$$
$$= (68 + 44 + 20 + 6) \times 10 = 138 \times 10 = \mathbf{1{,}380 \text{ mm}}$$

Highlighting in 3D: Deck 5 · 68 persons: 60 before OR-117, + 8 berths in the four new cabins.

Rev A needed 1,300 mm (N_1 was 60). The stair spans four decks nobody touched.

HOW IT WAS MEASURED

Clear width between handrails, from IFC geometry. Persons N from berths of the IfcSpaces each deck sends to Stair 3B.

IfcStair Stair-3B · checked Rev B · 14:32

Fix options · Accept with reason · Send to review · Zoom to

Fail F-07 — Rule engine

Stair 3B is narrower than the people it serves need

Stair 3B · Decks 4–7 · Deck 5 · FS Fire & life safety

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SonarIQ settings

Demo ship

- Units
- Rule packs
- AI endpoint
- Where data stays

UNITS

Default length unit: **mm** | in

- [x] Show a value in inches when its rule is written in inches
- [x] Lengths over 1 m in metres (4.2 m)

Format: `1,380 mm` · every value keeps its own unit toggle.

RULE PACKS

Fire & life safety · SOLAS II-2, FSS Code Ch.13	Verified	on
Accessibility · PVAG 2013 draft, ADA 2010	Verified	on
Class · DNV	To verify	on

AI ENDPOINT

- (•) On-premises (recommended)
- ( ) Approved cloud endpoint · your organisation's approved region
- ( ) Off · Analyse and evidence still work

`https://[your-endpoint]/v1` Test

Not tested yet. The AI explains, drafts edits and proposes fixes. It never decides a verdict.

WHERE THE DATA STAYS

Models, findings, decisions	Your SonarIQ server, on-premises
Signed history, BCF files	Same server · BCF also written next to the model
Sent to the AI endpoint	The question, the finding's rule text and numbers, element IDs
Never sent	Full model geometry, files, names in the signed history

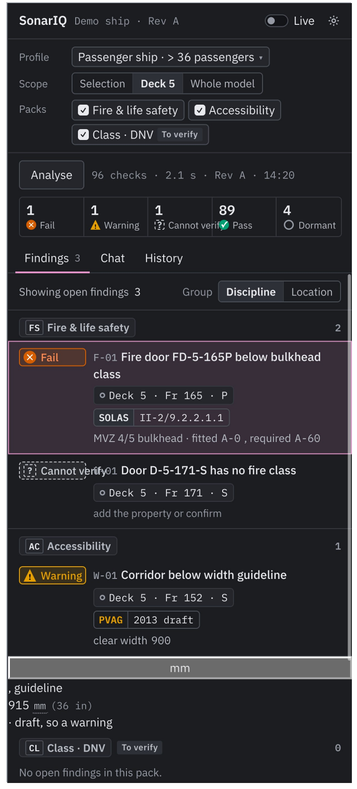
Cancel Save

SonarIQ · review board

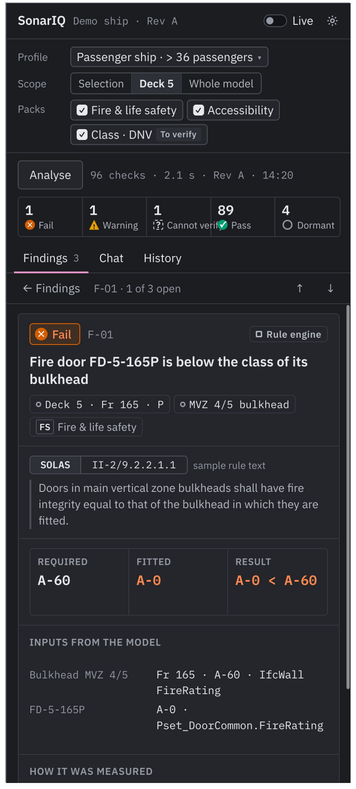
Analyse, evidence, accept with reason, inbox. Write on the board; there are no wrong answers.

Meyer Turku challenge · Since AI 2026
Demo ship · Deck 5 · Rev A / Rev B

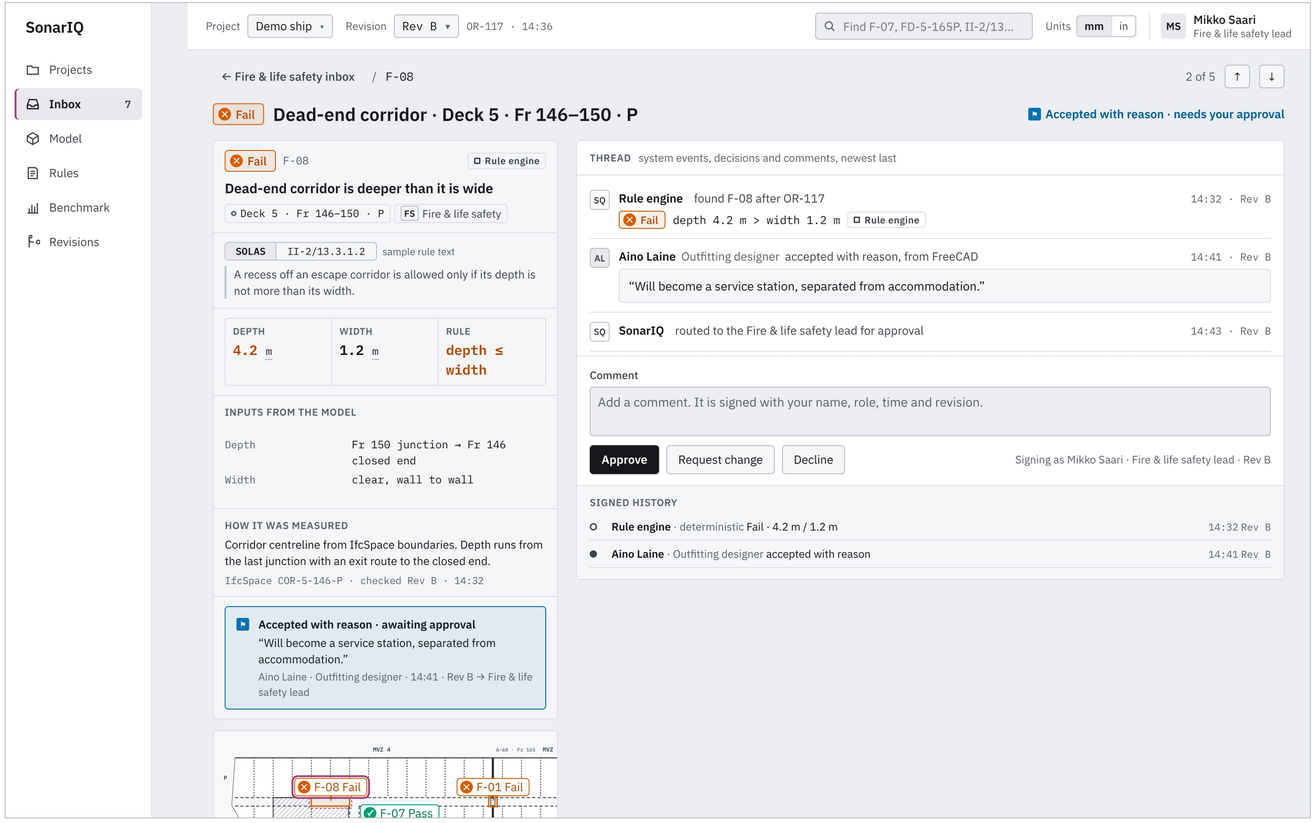
1 After Analyse P3



2 Evidence for F-01 P4



3 Accepted with reason, in the lead's inbox W3 · F-08



FIVE QUESTIONS

1 Are these the words you use?

Fail, Warning, Cannot verify, Dormant. Deck 5 · Fr 165 · P. MVZ 4/5. A-60.

2 Is this the evidence a reviewer needs?

Rule and clause, required vs fitted, inputs read from the model, how it was measured.

3 What is missing from the location?

Deck, frame, side, MVZ. Fire zone? Space number? Grid line?

4 Who approves an accepted finding?

Today: the Fire & life safety lead. Class? The owner? More than one person?

5 Which rule would you check first?

Pick the one you would trust least if a tool told you it passed.